

Logical Operators

Operator	Meaning
AND	Return TRUE if both component conditions are true.
OR	Returns TRUE is any of the component conditions is true.
NOT	Returns TRUE is the condition is FALSE.

- A logical combination combines the result of two component conditions to produce a single result.
- A row is returned only if the overall result of the condition is TRUE
- We can have multiple conditions in a WHERE clause using the logical operators.



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The AND logical operator.

- The AND operator will return TRUE only if both the component conditions are true.

AND	TRUE	FALSE	NULL
TRUE	TRUE	FALSE	NULL
FALSE	FALSE	FALSE	FALSE
NULL	NULL	FALSE	NULL

```
SELECT first_name, last_name, salary, department_id
FROM employees WHERE first_name = 'Steven' AND
salary=2200 ;
```



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```
SQL> SELECT first_name, last_name, salary, department_id
2 FROM employees
3 WHERE department_id = 80 AND salary > 6500;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
John	Russell	14000	80
Karen	Partners	13500	80
Alberto	Errazuriz	12000	80
Gerald	Cambrault	11000	80
Eleni	Zlotkey	10500	80
Peter	Tucker	15000	80
David	Bernstein	9500	80
Peter	Hall	9000	80
Christopher	Olsen	8000	80
Nanette	Cambrault	7500	80
Oliver	Tuvault	7000	80
Janette	King	10000	80
Patrick	Sully	9500	80
Allan	McEwen	9000	80
Lindsey	Smith	8000	80
Sarath	Sewall	7000	80
Clara	Vishnev	10500	80

```
SQL> SELECT first_name, last_name, salary, department_id
2 FROM employees
3 WHERE first_name = 'Steven';
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	King	24000	90
Steven	Markle	2200	50

```
SQL> SELECT first_name, last_name, salary, department_id
2 FROM employees
3 WHERE salary=2200;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	Markle	2200	50
Hazel	Philtanker	2200	50

```
SQL> SELECT first_name, last_name, salary, department_id
2 FROM employees
3 WHERE first_name = 'Steven' AND salary=2200
4 ;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	Markle	2200	50



The OR logical operator.



- The OR operator will return TRUE even if one of the component conditions are true.

OR	TRUE	FALSE	NULL
TRUE	TRUE	TRUE	TRUE
FALSE	TRUE	FALSE	NULL
NULL	TRUE	NULL	NULL

```
SELECT first_name, last_name, salary, department_id
FROM employees WHERE first_name = 'Steven' OR
salary=2200 ;
```



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```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
2 WHERE first_name = 'Steven';
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	King	24000	90
Steven	Markle	2200	50

```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
2 WHERE salary=2200;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	Markle	2200	50
Hazel	Philtanker	2200	50

```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
2 WHERE first_name = 'Steven' OR salary=2200
3 ;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	King	24000	90
Steven	Markle	2200	50
Hazel	Philtanker	2200	50

The NOT logical operator.

- The NOT operator IS used to negate the condition.

NOT	TRUE	FALSE	NULL
	FALSE	TRUE	NULL

```
SELECT first_name, last_name, salary, department_id
FROM employees WHERE department_id NOT IN
(50,60,70,80,90);
```

```
SELECT first_name, last_name, salary, department_id
FROM employees WHERE salary NOT BETWEEN 10000 AND
20000;
```



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```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
2 WHERE department_id NOT IN (50,60,80,90,100,110);
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Diana	Lorentz	4200	30
Den	Raphaely	11000	30
Alexander	Khoo	8200	30
Shelli	Baida	2900	30
Sigal	Tobias	2800	30
Guy	Himuro	2600	30
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Susan	Mavris	6000	40
Hermann	Baer	10000	70
NEW-EMP	EMP-3	10000	30
NEW-EMP	EMP-1	10000	30

13 rows selected.

Rules of Operator Precedence



Precedence	Operator Type
1	Arithmetic Operators
2	Concatenation Operator
3	Comparison Operator
4	IS [NOT] NULL, LIKE, [NOT] IN
5	[NOT] BETWEEN
6	Not Equal to
7	NOT Logical Condition
8	AND Logical Condition
9	OR Logical Condition



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Rules of Precedence



```
SELECT first_name, last_name, salary, department_id
FROM employees
WHERE first_name = 'Jennifer'
OR first_name = 'David'
AND salary='4800' ;
```

```
SELECT first_name, last_name, salary, department_id
FROM employees
WHERE (first_name = 'Jennifer'
OR first_name = 'David')
AND salary='4800' ;
```



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Sorting the result with ORDER BY Clause

- The ORDER BY clause can be used in the SELECT statements to sort the resultant rows in Ascending or Descending order.
ASC : Ascending order is the default.
DESC : Descending order
- The ORDER BY clause is always the last clause in a SELECT statement
- Use the NULLS FIRST or NULLS LAST to specify in which position the NULL values should be displayed..

```
SELECT first_name, last_name, salary, department_id
FROM employees WHERE department_id in (10,20,30)
ORDER BY department_ID;
```



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```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE department_id IN (10,20,30,40) OR department_id IS NULL
  3  ORDER BY department_id;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Shelli	Baida	2900	30
Sigal	Tobias	2800	30
Alexander	Khoo	8200	30
Den	Raphaely	11000	30
Diana	Lorentz	4200	30
Guy	Himuro	2600	30
Susan	Mavris	6000	40
Kimberely	Grant	7000	NULL

11 rows selected.

```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE department_id IN (10,20,30,40) OR department_id IS NULL
  3  ORDER BY department_id DESC;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Kimberely	Grant	7000	NULL
Susan	Mavris	6000	40
Alexander	Khoo	8200	30
Shelli	Baida	2900	30
Sigal	Tobias	2800	30
Den	Raphaely	11000	30
Diana	Lorentz	4200	30
Guy	Himuro	2600	30
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Jennifer	Whalen	2500	10

11 rows selected.

```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE department_id IN (10,20,30,40) OR department_id IS NULL
  3  ORDER BY department_id
  4  NULLS FIRST;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Kimberely	Grant	7000	NULL
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Sigal	Tobias	2800	30
Shelli	Baida	2900	30
Alexander	Khoo	8200	30
Den	Raphaely	11000	30
Diana	Lorentz	4200	30
Guy	Himuro	2600	30
Susan	Mavris	6000	40

11 rows selected.

```
SQL> SELECT first_name, last_name, salary, department_id FROM employees
2  WHERE department_id IN (10,20,30,40) OR department_id IS NULL
3  ORDER BY department_id DESC
4  NULLS LAST;
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Susan	Mavris	6000	40
Den	Raphaely	11000	30
Alexander	Khoo	8200	30
Shelli	Baida	2900	30
Sigal	Tobias	2800	30
Diana	Lorentz	4200	30
Guy	Himuro	2600	30
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Jennifer	Whalen	2500	10
Kimberely	Grant	7000	NULL

11 rows selected.



Points to Remember about the ORDER BY Clause.



- Numeric values are displayed with the lowest value first (1 to 999)
- Date values are displayed with the earliest date first (01-Jan-2021 before 01-Jan-2023)
- Character values are displayed in the alphabetical order. (A first and Z last)
- Null values are displayed last for ascending and first for descending sequence.
- Sorting can also be done by using the column which is not in the SELECT clause.
- **NOTE :** DESC in ORDER BY clause means Descending Order. It is different from the DESC <table_name> keyword used to describe a table.





Sorting by Alias

```
SELECT first_name, last_name, salary*12 annual_salary
from employees ORDER BY annual_salary;
```

Sorting in descending order

```
SELECT first_name, last_name, salary*12 annual_salary
from employees ORDER BY annual_salary desc;
```

Sorting by multiple columns

```
SELECT first_name, last_name, department_id,
salary*12 annual_salary from employees ORDER BY
department_id, annual_salary desc;
```

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```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
 2  WHERE department_id IN (10,20,30,40) OR department_id IS NULL
 3  ORDER BY deptid;
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Shelli	Baida	2900	30
Sigal	Tobias	2800	30
Alexander	Khoo	8200	30
Den	Raphaely	11000	30
Diana	Lorentz	4200	30
Guy	Himuro	2600	30
Susan	Mavris	6000	40
Kimberely	Grant	7000	NULL

11 rows selected.

```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
  2  WHERE department_id IN (10,20,30,40) OR department_id IS NULL
  3  ORDER BY 4 ;
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Shelli	Baida	2900	30
Sigal	Tobias	2800	30
Alexander	Khoo	8200	30
Den	Raphaely	11000	30
Diana	Lorentz	4200	30
Guy	Himuro	2600	30
Susan	Mavris	6000	40
Kimberely	Grant	7000	NULL

11 rows selected.

```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
  2  WHERE department_id IN (10,20,40,60,90)
  3  ORDER BY department_id , salary DESC ;
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Susan	Mavris	6000	40
Alexander	Hunold	12008	60
Bruce	Ernst	6000	60
David	Austin	4800	60
Valli	Pataballa	4800	60
Karen	Colmenares	2500	60
Steven	King	24000	90
Neena	Kochhar	17000	90
Lex	De_Haan	17000	90

12 rows selected.



- The Substitution Variables are used to store values temporarily with the Single-Ampersand (&) and Double-Ampersand (&&) variables.
- The Substitution Variables can be used to supplement
 - > WHERE conditions
 - > ORDER BY clause
 - > Column Expressions
 - > Table Names
 - > Entire SELECT statements.

Set the VERIFY [ON | OFF] to toggle the display of Substitution Variables.



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Using the Single-Ampersand '&' Substitution Variables



- The Substitution Variables '&' can be used to prompt the user for a input value for a variable.

```
SELECT employee_id,first_name,last_name, salary,  
department_id from employees where  
employee_id=&employee_id ;
```

- The Substitution Variables '&' can be used for Character and Date input values too.

```
SELECT employee_id,first_name,last_name, salary,  
department_id from employees where  
first_name=&first_name ;
```



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```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
  2  WHERE department_id = &deptid ;
Enter value for deptid: 10
old  2: WHERE department_id = &deptid
new  2: WHERE department_id = 10
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Jennifer	Whalen	2500	10

```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
  2  WHERE department_id = &deptid ;
Enter value for deptid: 20
old  2: WHERE department_id = &deptid
new  2: WHERE department_id = 20
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Michael	Hartstein	13000	20
Pat	Fay	6000	20

SQL>

```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
  2  WHERE &condition;
Enter value for condition: department_id = 10
old  2: WHERE &condition
new  2: WHERE department_id = 10
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Jennifer	Whalen	2500	10

```
SQL> SELECT first_name, last_name, salary, department_id deptid FROM employees
  2  WHERE &condition;
Enter value for condition: first_name = 'Steven'
old  2: WHERE &condition
new  2: WHERE first_name = 'Steven'
```

FIRST_NAME	LAST_NAME	SALARY	DEPTID
Steven	King	24000	90
Steven	Markle	2200	50

```
SQL> SELECT first_name, last_name, salary, department_id, &column5 FROM employees
  2  WHERE department_id = &deptid ;
Enter value for column5: hire_date
old  1: SELECT first_name, last_name, salary, department_id, &column5 FROM employees
new  1: SELECT first_name, last_name, salary, department_id, hire_date FROM employees
Enter value for deptid: 20
old  2: WHERE department_id = &deptid
new  2: WHERE department_id = 20
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	HIRE_DATE
Michael	Hartstein	13000	20	17-FEB-04
Pat	Fay	6000	20	17-AUG-05

```

SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE department_id IN (10,20,40)
  3  ORDER BY &column5;
Enter value for column5: department_id
old   3: ORDER BY &column5
new   3: ORDER BY department_id

```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Jennifer	Whalen	2500	10
Michael	Hartstein	13000	20
Pat	Fay	6000	20
Susan	Mavris	6000	40

```

SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE department_id IN (10,20,40)
  3  ORDER BY &column5;
Enter value for column5: salary
old   3: ORDER BY &column5
new   3: ORDER BY salary

```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Jennifer	Whalen	2500	10
Susan	Mavris	6000	40
Pat	Fay	6000	20
Michael	Hartstein	13000	20

```

SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE first_name = &first_name;
Enter value for first_name: Steven
old   2: WHERE first_name = &first_name
new   2: WHERE first_name = Steven
WHERE first_name = Steven
          *

```

```

ERROR at line 2:
ORA-00904: "STEVEN": invalid identifier

```

```

SQL> SELECT first_name, last_name, salary, department_id FROM employees
  2  WHERE first_name = '&first_name';
Enter value for first_name: Steven
old   2: WHERE first_name = '&first_name'
new   2: WHERE first_name = 'Steven'

```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
Steven	King	24000	90
Steven	Markle	2200	50

```
SQL> █
```

- The Substitution Variables '&' can be used if we want to use the same variable more than once without prompting the user for the input multiple times.

```
SELECT employee_id,first_name,last_name, &&Column3
FROM employees
ORDER BY &Column3;
```



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```
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
2  WHERE department_id IN (10,20,40)
3  ORDER BY &column5;
Enter value for column5: hire_date
old 1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new 1: SELECT first_name, last_name, salary, department_id,hire_date FROM employees
Enter value for column5: hire_date
old 3: ORDER BY &column5
new 3: ORDER BY hire_date
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	HIRE_DATE
Susan	Mavris	6000	40	07-JUN-02
Jennifer	Whalen	2500	10	17-SEP-03
Michael	Hartstein	13000	20	17-FEB-04
Pat	Fay	6000	20	17-AUG-05

```
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
2  WHERE department_id IN (10,20,40)
3  ORDER BY &column5;
Enter value for column5: hire_date
old 1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new 1: SELECT first_name, last_name, salary, department_id,hire_date FROM employees
Enter value for column5: job_id
old 3: ORDER BY &column5
new 3: ORDER BY job_id
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	HIRE_DATE
Jennifer	Whalen	2500	10	17-SEP-03
Susan	Mavris	6000	40	07-JUN-02
Michael	Hartstein	13000	20	17-FEB-04
Pat	Fay	6000	20	17-AUG-05

```
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
2 WHERE department_id IN (10,20,40)
3 ORDER BY &column5;
Enter value for column5: hire_date
old 1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new 1: SELECT first_name, last_name, salary, department_id,hire_date FROM employees
old 3: ORDER BY &column5
new 3: ORDER BY hire_date
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	HIRE_DATE
Susan	Mavris	6000	40	07-JUN-02
Jennifer	Whalen	2500	10	17-SEP-03
Michael	Hartstein	13000	20	17-FEB-04
Pat	Fay	6000	20	17-AUG-05

```
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
2 WHERE department_id IN (10,20,40)
3 ORDER BY &column5;
old 1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new 1: SELECT first_name, last_name, salary, department_id,hire_date FROM employees
old 3: ORDER BY &column5
new 3: ORDER BY hire_date
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	HIRE_DATE
Susan	Mavris	6000	40	07-JUN-02
Jennifer	Whalen	2500	10	17-SEP-03
Michael	Hartstein	13000	20	17-FEB-04
Pat	Fay	6000	20	17-AUG-05

```
SQL> undefine column5
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
2 WHERE department_id IN (10,20,40)
3 ORDER BY &column5;
Enter value for column5: job_id
old 1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new 1: SELECT first_name, last_name, salary, department_id,job_id FROM employees
old 3: ORDER BY &column5
new 3: ORDER BY job_id
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	JOB_ID
Jennifer	Whalen	2500	10	AD_ASST
Susan	Mavris	6000	40	HR_REP
Michael	Hartstein	13000	20	MK_MAN
Pat	Fay	6000	20	MK_REP

```
SQL> undefine column5
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
2 WHERE department_id IN (10,20,40)
3 ORDER BY &column5;
Enter value for column5: job_id
old 1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new 1: SELECT first_name, last_name, salary, department_id,job_id FROM employees
old 3: ORDER BY &column5
new 3: ORDER BY job_id
```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	JOB_ID
Jennifer	Whalen	2500	10	AD_ASST
Susan	Mavris	6000	40	HR_REP
Michael	Hartstein	13000	20	MK_MAN
Pat	Fay	6000	20	MK_REP

```

SQL> undefine column5
SQL> define column5 = job_id
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
  2  WHERE department_id IN (10,20,40)
  3  ORDER BY &column5;
old  1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new  1: SELECT first_name, last_name, salary, department_id,job_id FROM employees
old  3: ORDER BY &column5
new  3: ORDER BY job_id

```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	JOB_ID
Jennifer	Whalen	2500	10	AD_ASST
Susan	Mavris	6000	40	HR_REP
Michael	Hartstein	13000	20	MK_MAN
Pat	Fay	6000	20	MK_REP

```

SQL> ACCEPT column5 PROMPT 'Kindly Enter the Value for Column5 : '
Kindly Enter the Value for Column5 : job_id
SQL> SELECT first_name, last_name, salary, department_id,&column5 FROM employees
  2  WHERE department_id IN (10,20,40)
  3  ORDER BY &column5;
old  1: SELECT first_name, last_name, salary, department_id,&column5 FROM employees
new  1: SELECT first_name, last_name, salary, department_id,job_id FROM employees
old  3: ORDER BY &column5
new  3: ORDER BY job_id

```

FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID	JOB_ID
Jennifer	Whalen	2500	10	AD_ASST
Susan	Mavris	6000	40	HR_REP
Michael	Hartstein	13000	20	MK_MAN
Pat	Fay	6000	20	MK_REP